KiCad DRU

PROVES chiplet footprints keep their courtyard clearance, at the moment of the drag

BLIND TO pillars, die artwork and z, none of which exist yet

operates on footprint courtyards, not on the manifest bounding box

KLayout assembly DRC (`run_drc.py`)

PROVES the drawn geometry satisfies the ASM and IXN rules

BLIND TO z, and geometry that was never drawn

`chiplet2dbx` plus `check_3dblox`

PROVES z, stacking and bump alignment are internally consistent; the only stage with a z axis

BLIND TO mask artwork, routing, anything only in the GDS

Chiplet Export plugin

`hyp_to_gds.py`

`chiplet2dbx.py`

KiCad board
`.kicad_pcb`

→

`<board>.hyp`
plus intermediate
`.chiplet`

→

assembly GDS
plus **`.boundaries.json`**, **`.pillars.json`**, finalised **`.chiplet`**

→

3Dblox model
`.3dbx`

producer-side checks inside `hyp_to_gds`

Cu-pillar DRC per die at generation time

refuses malformed output: unmapped routing layers, no die GDS loaded, unreachable pillar generator, unfinalised **`.chiplet`**

not substitutes for the four stages; they fail earlier and more specifically

`pads_vs_pillars`

PROVES every die pad has a pillar under it, by name and in frame

BLIND TO whether those pillars are legally spaced

a clean result at one stage does not imply a clean result at another